Topic: Convolution integrals

Question: Find the convolution of $f(t) = \sin t$ and $g(t) = e^{-t}$.

Answer choices:

A $f(t) * g(t) = \frac{1}{2}(e^{-t} + \sin t - \cos t)$

B $f(t) * g(t) = \frac{1}{2}(e^{-t} - \sin t - \cos t)$

C $f(t) * g(t) = e^{-t} - 2\sin t$

D $f(t) * g(t) = e^{-t} - 2\cos t$

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Solution: A

To find the convolution, we'll first find $f(\tau)$ and $g(t-\tau)$, the two functions that we need for our convolution integral.

$$f(\tau) = \sin \tau$$

$$g(t-\tau) = e^{-(t-\tau)} = e^{\tau - t}$$

So the convolution integral gives

$$f(t) * g(t) = \int_0^t f(\tau) g(t-\tau)\ d\tau$$

$$f(t) * g(t) = \int_0^t \sin \tau (e^{\tau - t})\ d\tau$$

$$f(t) * g(t) = \int_0^t \sin \tau (e^{-t} e^{\tau})\ d\tau$$

$$f(t) * g(t) = e^{-t} \int_0^t e^{\tau} \sin \tau\ d\tau$$

We'll use integration by parts to evaluate the integral, letting

$$u = \sin \tau \qquad dv = e^{\tau}\ d\tau$$

$$du = \cos \tau\ d\tau \qquad v = e^{\tau}$$

Then the convolution is given by

$$f(t) * g(t) = e^{-t} \left[uv \Big|_0^t - \int_0^t v\ du \right]$$

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$$f(t) * g(t) = e^{-t}\left[e^{\tau}\sin\tau\Big|_0^t - \int_0^t e^{\tau}\cos\tau\ d\tau\right]$$

We'll use integration by parts to evaluate the integral, letting

$$u = \cos\tau \qquad dv = e^{\tau}\ d\tau$$

$$du = -\sin\tau\ d\tau \qquad v = e^{\tau}$$

Then the convolution is given by

$$f(t) * g(t) = e^{-t}\left[e^{\tau}\sin\tau\Big|_0^t - \left(e^{\tau}\cos\tau\Big|_0^t + \int_0^t e^{\tau}\sin\tau\ d\tau\right)\right]$$

$$f(t) * g(t) = e^{-t}\left(e^{\tau}\sin\tau\Big|_0^t - e^{\tau}\cos\tau\Big|_0^t - \int_0^t e^{\tau}\sin\tau\ d\tau\right)$$

Integrate the remaining function, and evaluate over the interval.

$$f(t) * g(t) = e^{-t}\left(e^{t}\sin t - e^{t}\cos t + 1 - \int_0^t e^{\tau}\sin\tau\ d\tau\right)$$

$$f(t) * g(t) = \sin t - \cos t + e^{-t} - e^{-t}\int_0^t e^{\tau}\sin\tau\ d\tau$$

$$e^{-t}\int_0^t e^{\tau}\sin\tau\ d\tau = \sin t - \cos t + e^{-t} - e^{-t}\int_0^t e^{\tau}\sin\tau\ d\tau$$

$$2e^{-t}\int_0^t e^{\tau}\sin\tau\ d\tau = \sin t - \cos t + e^{-t}$$

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$$e^{-t}\int_0^t e^{\tau}\sin\tau\ d\tau = \frac{\sin t - \cos t + e^{-t}}{2}$$

$$f(t) * g(t) = \sin t - \cos t + e^{-t} - \frac{\sin t - \cos t + e^{-t}}{2}$$

$$f(t) * g(t) = \frac{\sin t - \cos t + e^{-t}}{2}$$

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Topic: Convolution integrals

Question: Find the convolution of $f(t) = t^3$ and $g(t) = e^t$.

Answer choices:

A $f(t) * g(t) = -t^3 + t + e^t$

B $f(t) * g(t) = -t^3 - 3t^2 - 6t - 6 + 6e^t$

C $f(t) * g(t) = -t^3 + t^2 - t + e^t$

D $f(t) * g(t) = -3t^3 - 3t^2 - t - 1 + 6e^t$

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Solution: B

To find the convolution, we'll first find $f(\tau)$ and $g(t-\tau)$, the two functions that we need for our convolution integral.

$$f(\tau) = \tau^3$$

$$g(t-\tau) = e^{t-\tau}$$

So the convolution integral gives

$$f(t) * g(t) = \int_0^t f(\tau)g(t-\tau)\ d\tau$$

$$f(t) * g(t) = \int_0^t \tau^3 e^{t-\tau}\ d\tau$$

$$f(t) * g(t) = \int_0^t \tau^3 e^t e^{-\tau}\ d\tau$$

$$f(t) * g(t) = e^t \int_0^t \tau^3 e^{-\tau}\ d\tau$$

We'll use integration by parts to evaluate the integral, letting

$$u = \tau^3 \qquad dv = e^{-\tau}\ d\tau$$

$$du = 3\tau^2\ d\tau \qquad v = -e^{-\tau}$$

Then the convolution is given by

$$f(t) * g(t) = e^t \left[uv\Big|_0^t - \int_0^t v\ du \right]$$

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$$f(t) * g(t) = e^t \left[-\tau^3 e^{-\tau} \Big|_0^t + 3 \int_0^t e^{-\tau} \tau^2 \, d\tau \right]$$

We'll use integration by parts to evaluate the integral, letting

$$u = \tau^2 \qquad dv = e^{-\tau} \, d\tau$$

$$du = 2\tau \, d\tau \qquad v = -e^{-\tau}$$

Then the convolution is given by

$$f(t) * g(t) = e^t \left[-\tau^3 e^{-\tau} \Big|_0^t + 3 \left(-\tau^2 e^{-\tau} \Big|_0^t + 2 \int_0^t \tau e^{-\tau} \, d\tau \right) \right]$$

$$f(t) * g(t) = e^t \left[-\tau^3 e^{-\tau} \Big|_0^t - 3\tau^2 e^{-\tau} \Big|_0^t + 6 \int_0^t \tau e^{-\tau} \, d\tau \right]$$

We'll use integration by parts to evaluate the integral, letting

$$u = \tau \qquad dv = e^{-\tau} \, d\tau$$

$$du = d\tau \qquad v = -e^{-\tau}$$

Then the convolution is given by

$$f(t) * g(t) = e^t \left[-\tau^3 e^{-\tau} \Big|_0^t - 3\tau^2 e^{-\tau} \Big|_0^t + 6 \left(-\tau e^{-\tau} \Big|_0^t + \int_0^t e^{-\tau} \, d\tau \right) \right]$$

$$f(t) * g(t) = e^t \left(-\tau^3 e^{-\tau} \Big|_0^t - 3\tau^2 e^{-\tau} \Big|_0^t - 6\tau e^{-\tau} \Big|_0^t + 6 \int_0^t e^{-\tau} \, d\tau \right)$$

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$$f(t) * g(t) = e^t \left(-\tau^3 e^{-\tau} - 3\tau^2 e^{-\tau} - 6\tau e^{-\tau} - 6e^{-\tau} \Big|_0^t \right)$$

Integrate the remaining function, and evaluate over the interval.

$$f(t) * g(t) = e^t(-t^3 e^{-t} - 3t^2 e^{-t} - 6te^{-t} - 6e^{-t} + 6)$$

$$f(t) * g(t) = -t^3 - 3t^2 - 6t - 6 + 6e^t$$

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Topic: Convolution integrals

Question: Use a convolution integral to find the inverse transform.

$$H(s) = \left(\frac{s}{s^2+1}\right)\left(\frac{1}{(s-1)^2}\right)$$

Answer choices:

A $f(t) * g(t) = te^t - \cos t$

B $f(t) * g(t) = e^{2t} + t \sin t$

C $f(t) * g(t) = \frac{1}{2}(te^t - \sin t)$

D $f(t) * g(t) = e^t - \cos t$

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Solution: C

In the function $H(s)$, we have

$$F(s) = \frac{s}{s^2 + 1}$$

$$G(s) = \frac{1}{(s-1)^2}$$

Using the Laplace transforms

$$\mathscr{L}(\cos(at)) = \frac{s}{s^2 + a^2}$$

$$\mathscr{L}(t^n e^{at}) = \frac{n!}{(s-a)^{n+1}}$$

we get

$$f(t) = \cos t$$

$$g(t) = te^t$$

To find the convolution, we'll first find $f(\tau)$ and $g(t-\tau)$, the two functions that we need for our convolution integral.

$$f(\tau) = \cos \tau$$

$$g(t-\tau) = (t-\tau)e^{t-\tau}$$

So the convolution integral gives

$$f(t) * g(t) = \int_0^t f(\tau)g(t-\tau)\ d\tau$$

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$$f(t) * g(t) = \int_0^t (t-\tau)(\cos\tau)e^{t-\tau}\ d\tau$$

$$f(t) * g(t) = \int_0^t (t\cos\tau)e^{t-\tau}\ d\tau - \int_0^t (\tau\cos\tau)e^{t-\tau}\ d\tau$$

$$f(t) * g(t) = te^t\int_0^t e^{-\tau}\cos\tau\ d\tau - e^t\int_0^t \tau e^{-\tau}\cos\tau\ d\tau$$

We'll use integration by parts to evaluate the second integral, letting

$$u = \tau \qquad dv = e^{-\tau}\cos\tau\ d\tau$$

$$du = d\tau \qquad v = \frac{1}{2}e^{-\tau}(\sin\tau - \cos\tau)$$

Then the convolution is given by

$$\int_0^t \tau e^{-\tau}\cos\tau\ d\tau = uv\Big|_0^t - \int_0^t v\ du$$

$$\int_0^t \tau e^{-\tau}\cos\tau\ d\tau = \frac{1}{2}\tau e^{-\tau}(\sin\tau - \cos\tau)\Big|_0^t - \int_0^t \frac{1}{2}e^{-\tau}(\sin\tau - \cos\tau)\ d\tau$$

$$\int_0^t \tau e^{-\tau}\cos\tau\ d\tau = \frac{1}{2}\tau e^{-\tau}(\sin\tau - \cos\tau)\Big|_0^t - \frac{1}{2}\int_0^t \frac{d}{d\tau}(-e^{-\tau}\sin\tau)\ d\tau$$

$$\int_0^t \tau e^{-\tau}\cos\tau\ d\tau = \frac{1}{2}te^{-t}(\sin t - \cos t) + \frac{1}{2}e^{-\tau}\sin\tau\Big|_0^t$$

$$\int_0^t \tau e^{-\tau}\cos\tau\ d\tau = \frac{1}{2}te^{-t}(\sin t - \cos t) + \frac{1}{2}e^{-t}\sin t$$

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$$\int_0^t \tau e^{-\tau} \cos\tau \, d\tau = \frac{e^{-t}}{2}(t \sin t - t \cos t + \sin t)$$

We'll use integration by parts to evaluate the integral first integral, letting

$$u = \cos\tau \qquad dv = e^{-\tau} \, d\tau$$

$$du = -\sin\tau \, d\tau \qquad v = -e^{-\tau}$$

Then the convolution is given by

$$\int_0^t e^{-\tau} \cos\tau \, d\tau = uv\Big|_0^t - \int_0^t v \, du$$

$$\int_0^t e^{-\tau} \cos\tau \, d\tau = -e^{-\tau} \cos\tau\Big|_0^t - \int_0^t e^{-\tau} \sin\tau \, d\tau$$

We'll use integration by parts to evaluate the integral this remaining integral, letting

$$u = \sin\tau \qquad dv = e^{-\tau} \, d\tau$$

$$du = \cos\tau \, d\tau \qquad v = -e^{-\tau}$$

Then the value of the integral is

$$\int_0^t e^{-\tau} \cos\tau \, d\tau = -e^{-\tau} \cos\tau\Big|_0^t - \left(-e^{-\tau} \sin\tau + \int_0^t e^{-\tau} \cos\tau \, d\tau \right)$$

$$\int_0^t e^{-\tau} \cos\tau \, d\tau = -e^{-\tau} \cos\tau\Big|_0^t + e^{-\tau} \sin\tau\Big|_0^t - \int_0^t e^{-\tau} \cos\tau \, d\tau$$

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$$2\int_0^t e^{-\tau}\cos\tau\ d\tau = -e^{-\tau}\cos\tau\Big|_0^t + e^{-\tau}\sin\tau\Big|_0^t$$

$$2\int_0^t e^{-\tau}\cos\tau\ d\tau = -e^{-t}\cos t + 1 + e^{-t}\sin t$$

$$\int_0^t e^{-\tau}\cos\tau\ d\tau = \frac{e^{-t}\sin t - e^{-t}\cos t + 1}{2}$$

Integrate the remaining function, and evaluate over the interval.

$$f(t) * g(t) = te^t\left(\frac{e^{-t}\sin t - e^{-t}\cos t + 1}{2}\right) - e^t\left(\frac{e^{-t}}{2}(t\sin t - t\cos t + \sin t)\right)$$

$$f(t) * g(t) = \frac{t\sin t - t\cos t + te^t}{2} - \frac{t\sin t - t\cos t + \sin t}{2}$$

$$f(t) * g(t) = \frac{te^t - \sin t}{2}$$

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